percent of light speed, for example, weigh a lot more than their stationary cousins. In fact, they are about 22 times as heavy—literally. (The masses recorded in Table 1.1 are for particles at rest.) But the more massive an object is, the harder it is to increase its speed. Pushing a child on a bicycle is one thing, pushing a Mack truck is quite another. So, as a muon moves more quickly it gets ever more difficult to further increase its speed. At 99.999 percent of light speed the mass of a muon has increased by a factor of 224; at 99.99999999 percent of light speed it has increased by a factor of more than 70,000. Since the mass of the muon increases without limit as its speed approaches that of light, it would require a push with an infinite amount of energy to reach or to cross the light barrier. This, of course, is impossible and hence absolutely nothing can travel faster than the speed of light.

As we shall see in the next chapter, this conclusion plants the seeds for the second major conflict faced by physics during the past century and ultimately spells doom for another venerable and cherished theory —Newton’s universal theory of gravity.